
**Lab Report 1: Design and Analysis of CMOS Inverter circuit in
Cadence Virtuoso.**



Department of Electrical and Electronic Engineering

Course Title: VLSI Design and Testing Lab

Course code: EEE 356

Group No: 12

Submitted By:

Name: Prapto Debnath

Registration No: 2021338016

Name: Sujon Bepary

Registration No: 2021338044

Dept. of Electrical & Electronic Engineering

Submitted To:

Myen Uddin

Lecturer

Department of Electrical & Electronic Engineering,
Shahjalal University of Science and Technology, Sylhet.

Date of Submission

15-08-2026

Theory

A CMOS (Complementary Metal-Oxide-Semiconductor) inverter is a fundamental building block in digital electronics and integrated circuits. It consists of both NMOS (N-type Metal-Oxide-Semiconductor) and PMOS (P-type Metal-Oxide-Semiconductor) transistors configured in a complementary manner. The term “complementary” refers to the use of both types of transistors to achieve low power consumption and other desirable characteristics.

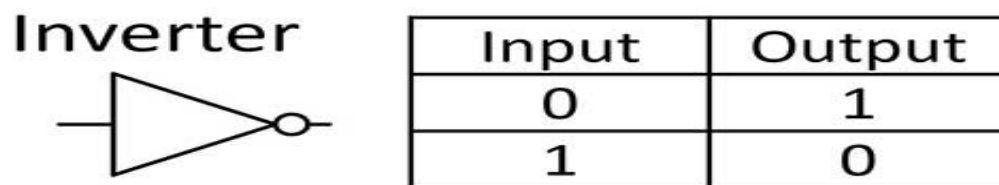


Figure: Inverter Truth Table

CMOS INVERTER CIRCUIT

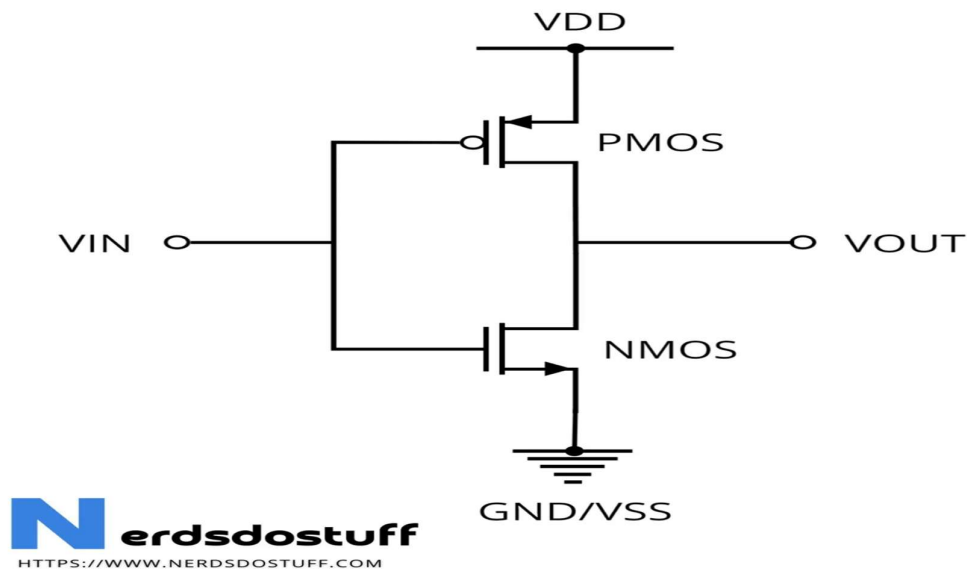


Figure: CMOS Inverter Circuit

Ref: <https://nerdsdostuff.com/wp-content/uploads/2024/03/CMOS-Inverter-Circuit-Image-1295x1536.png>

1. Input State (High Voltage – Logic 1):

When the input voltage (V_{IN}) is at a high level (logic 1), the PMOS transistor conducts, while the NMOS transistor is in the off state. The high input voltage turns ON the PMOS transistor because a positive voltage at its gate relative to its source enhances the conductive channel, allowing current to flow from the power supply (V_{DD}) to the output (V_{OUT}).

At the same time, the NMOS transistor is OFF since a high voltage at its gate relative to its source would create a reverse-biased condition, blocking the flow of current from the output to the ground (V_{SS}).

2. Output State (Low Voltage – Logic 0):

With the PMOS transistor conducting, the output voltage (V_{OUT}) is pulled up towards the power supply voltage (V_{DD}). This creates a logic low (0) at the output because the output voltage is close to V_{DD} .

The NMOS transistor remains OFF, so there is no current flowing from the output to the ground. The NMOS transistor effectively isolates the output from the ground, maintaining a high voltage.

3. Input State (Low Voltage – Logic 0):

When the input voltage (V_{IN}) is at a low level (logic 0), the NMOS transistor conducts, and the PMOS transistor is in the off state.

The low input voltage turns ON the NMOS transistor because a positive voltage at its gate relative to its source allows current to flow from the output to the ground (V_{SS}).

Simultaneously, the PMOS transistor is OFF since a low voltage at its gate relative to its source creates a reverse-biased condition, blocking the flow of current from the power supply to the output.

4. Output State (High Voltage – Logic 1):

With the NMOS transistor conducting, the output voltage (V_{OUT}) is pulled down towards the ground. This creates a logic high (1) at the output because the output voltage is close to V_{SS} .

The PMOS transistor remains OFF, so there is no current flowing from the power supply to the output. The PMOS transistor effectively isolates the output from the power supply, maintaining a low voltage.

Lab Schematics:

A transient analysis was performed in Cadence Virtuoso to verify the logical operation of the inverter. The input signal was configured as a periodic digital waveform alternating between approximately 0 V and 1.2 V. The transient simulation was observed over a time interval of approximately 200 ns.

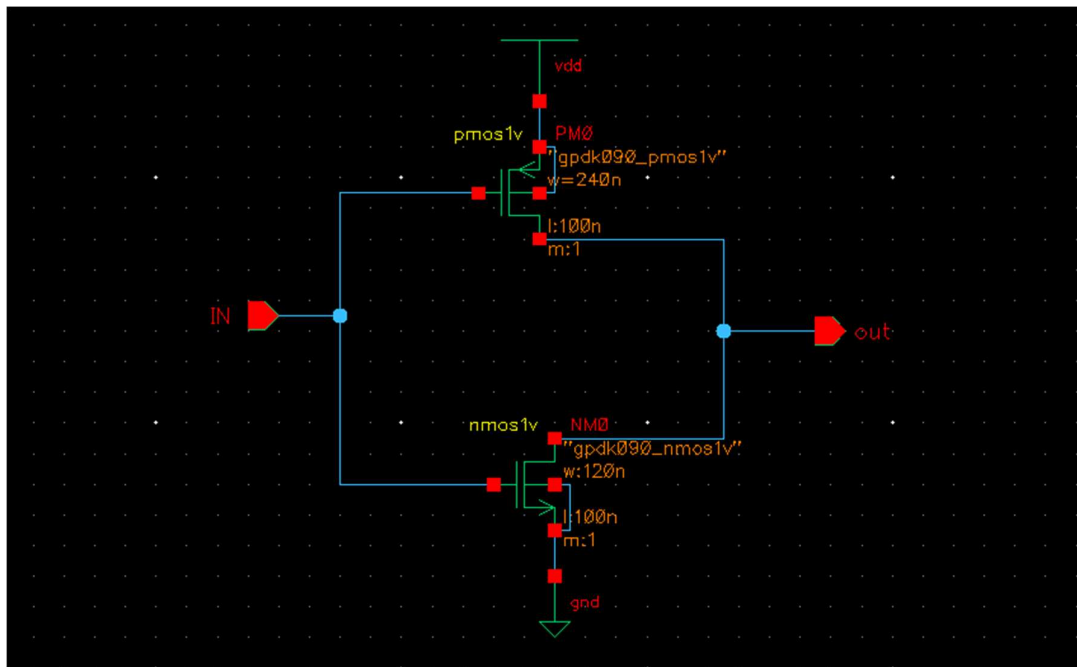


Figure : Circuit schematic of the experimental setup

Simulation Results:

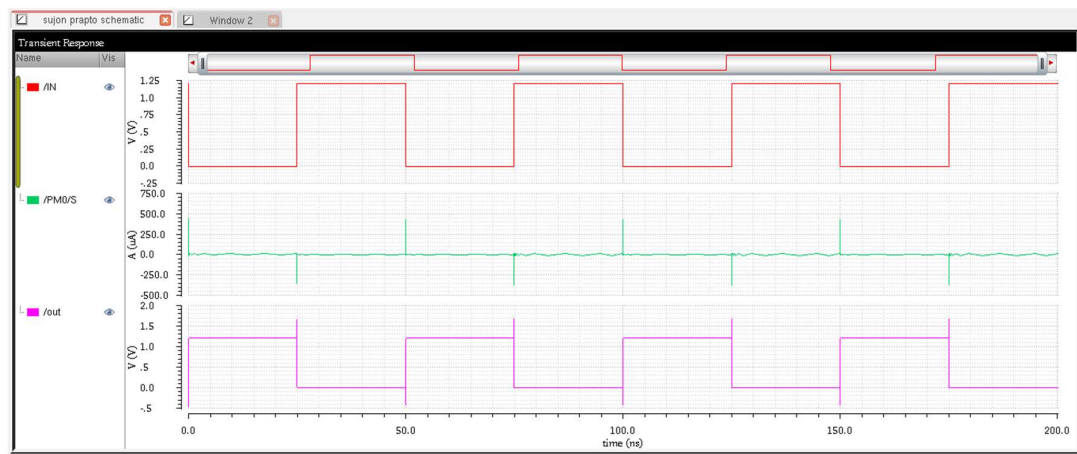


Figure : Input and Output Plot of the experimental setup

The simulation also shows short current spikes during the transitions of the input signal.

These current peaks occur because both the PMOS and NMOS can conduct simultaneously for a short period during switching. Consequently, a temporary current path exists between Vdd and ground. The current response in the simulation is close to zero during the stable logic states, with pronounced positive or negative transient peaks occurring around the switching instants. This behavior is characteristic of CMOS circuits and demonstrates that the major power consumption of the inverter occurs during switching rather than during steady-state operation.

Discussion:

The Cadence Virtuoso simulation successfully demonstrates the operation of the designed CMOS inverter.

When the input signal is LOW, the NMOS transistor remains OFF because its gate-to-source voltage is below its threshold voltage. At the same time, the PMOS transistor is ON and connects the output to Vdd. Therefore, the output rises to approximately 1.2 V, representing logic HIGH.

When the input signal becomes HIGH, the PMOS turns OFF and the NMOS turns ON. The NMOS then establishes a low-resistance path between the output and ground, causing the output voltage to fall close to 0 V. Hence, the circuit correctly performs logical inversion.

The simulation also demonstrates transient current peaks during the output transitions. These peaks result from the charging and discharging of parasitic/load capacitances as well as the temporary simultaneous conduction of the PMOS and NMOS devices.

The PMOS width was selected as 240 nm, twice the NMOS width of 120 nm. This is appropriate because hole mobility in a PMOS is lower than electron mobility in an NMOS. Increasing the PMOS width increases its drive strength and helps make the rising and falling transitions more balanced.

The output waveform shown in the simulation reaches approximately the supply voltage for the HIGH state and approaches ground for the LOW state. This indicates that the inverter provides good logic-level restoration.

A small amount of waveform transition delay is expected because the MOSFETs have finite resistance and the circuit contains parasitic capacitances. The output therefore cannot change instantaneously when the input changes. The propagation delay is associated with charging and discharging of the output capacitance.

The simulation confirms the fundamental advantages of CMOS logic:

- i. Full-swing logic levels
- ii. Very low steady-state current
- iii. High noise immunity
- iv. Low static power consumption
- v. Good switching capability

Removing the spike

To remove the spike, we need to increase the rise time, the fall time, and the delay time. we can use the following configurations.

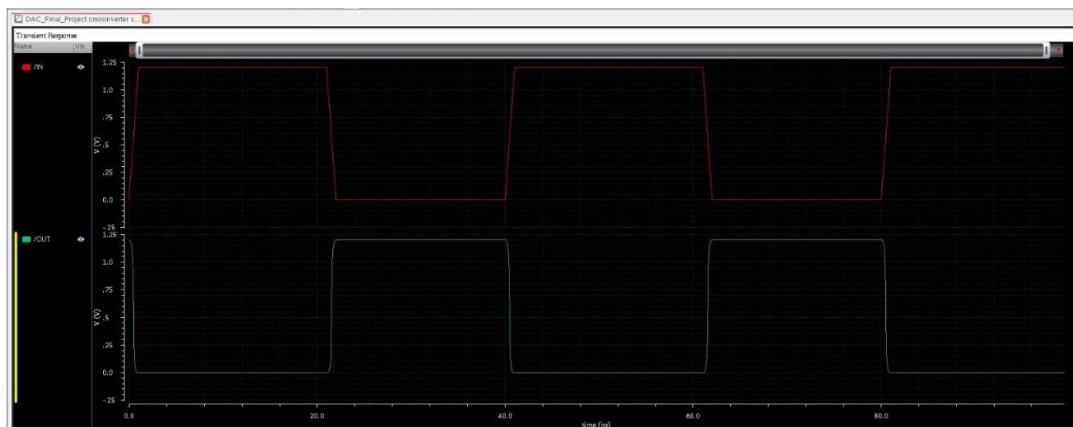


Figure : Spike removed plot.

**Lab Report 2: DC Characteristics of CMOS Inverter & I-V
Characteristics of nMOS**



Department of Electrical and Electronic Engineering

Course Title: VLSI Design and Testing Lab

Course code: EEE 356

Group No: 12

Submitted By:

Name: Prapto Debnath

Registration No: 2021338016

Name: Sujon Bepary

Registration No: 2021338044

Dept. of Electrical & Electronic Engineering

Submitted To:

Myen Uddin

Lecturer

Department of Electrical & Electronic Engineering,
Shahjalal University of Science and Technology, Sylhet.

Date of Submission

15-08-2026

Theory:

A CMOS inverter is a basic digital circuit consisting of a pMOS transistor connected to V_{dd} and an nMOS transistor connected to ground. Their gates are connected together as the input, and their drains are connected together as the output.

The DC characteristics describe the relationship between the input voltage and output voltage when the circuit is in steady state. This relationship is called the Voltage Transfer Characteristic (VTC).

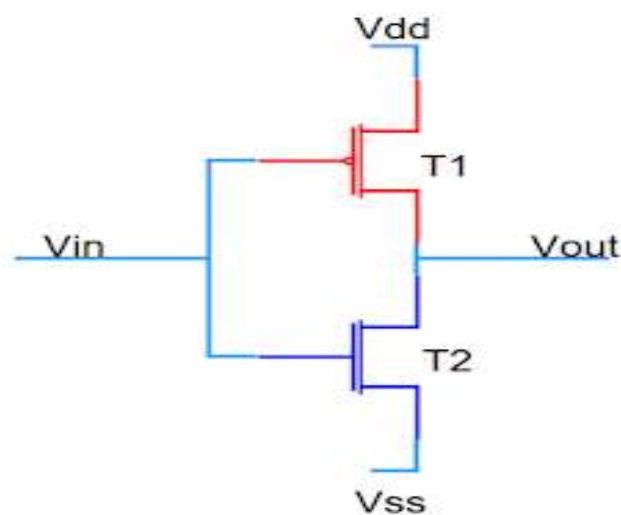


Figure-1

Figure-1 shows the schematic of a CMOS inverter. As we can see it has two transistors: a pull-up pMOS transistor (T1) and a pull-down nMOS transistor (T2). When the input voltage V_{in} is equal to V_{dd} , we get an output voltage of V_{ss} (mostly equal to 0) and vice versa. Since it inverts the logic level of input, this circuit is called an inverter. Figure-2 shows the voltage transfer characteristics (VTC) or DC characteristics of an inverter.

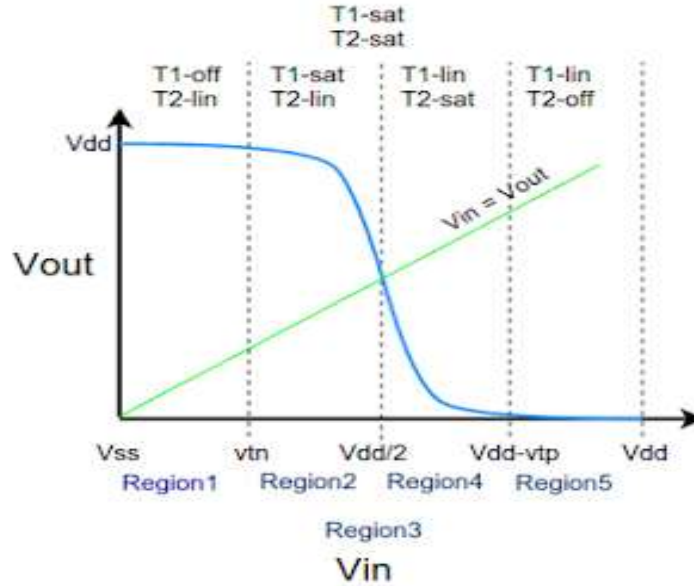


Figure-2

The VTC is divided into five regions (1-5) for easy of understanding

Region-1

In this region the input is in the range of $(0, V_{tn})$. Since the input voltage is less than V_{tn} , the NMOS is in cutoff region. No current flows from V_{dd} to V_{ss} , The entire V_{dd} will appear at the Output terminal.

- NMOS is in cutoff as $V_{gs} < V_{tn}$
- PMOS is in linear as $V_{gsp} < V_{tp}$ and $V_{dsp} > V_{gsp} - V_{tp}$.
- Zero current flows from supply voltage and the power dissipation is zero.

Region-2

In this region the input is in the range of $(V_{tn}, V_{dd}/2)$. Since the input voltage is greater than V_{tn} the NMOS is conducting and it jumps to saturation as it has large V_{ds} across it (V_{out} is high). PMOS still remains in the linear region.

- NMOS is in saturation as $V_{gs} > V_{tn}$ and $V_{out} > V_{in} - V_{tn}$.
- PMOS is in linear region as $V_{dsp} > V_{gsp} - V_{tp}$.
- since both the transistors are conducting some amount of current flows from supply in this region.

Region-3

In this region the input voltage is $V_{dd}/2$. At this point the output voltage is also $V_{dd}/2$ as one can see in figure-2. At this voltage both the NMOS and PMOS are in saturation and the output drops drastically from V_{dd} to $V_{dd}/2$. At this point a large amount of current flows from the supply. Most of the power consumed in CMOS inverter is at this point. So care should be taken that the Input should not stay at $V_{dd}/2$ for more amount of time.

- NMOS is in saturation as $V_{gs} > V_{tn}$ and $V_{out} > V_{in} - V_{tn}$.
- PMOS is in saturation as $V_{gsp} < V_{tp}$ and $V_{dsp} < V_{gsp} - V_{tp}$.
- Large amount of current is drawn from supply and hence large power dissipation.

Region-4

In this region the input voltage is in the range of $(V_{dd}/2, V_{dd}-V_{tp})$. Here the PMOS remains in saturation as $V_{out} < V_{in} - V_{tp}$ and $V_{gsp} < V_{tp}$. But the NMOS moves from saturation to linear region since the drain to source voltage now is less than $V_{gsn} - V_{tn}$.

- NMOS is in linear as $V_{gs} > V_{tn}$ and $V_{out} < V_{in} - V_{tn}$.
- PMOS is in saturation as $V_{gsp} < V_{tp}$ and $V_{dsp} < V_{gsp} - V_{tp}$.
- A medium amount of current is drawn as NMOS is in linear region and power dissipation is low.

Region-5

In this region the input voltage is in the range of $(V_{dd}-V_{tp}, V_{dd})$. Here the PMOS moves from saturation to cutoff as the V_{gsp} is so high that $V_{gsp} > V_{tp}$. The NMOS still remains in linear as the drain to source voltage now is less than $V_{gsn} - V_{tn}$.

- NMOS is in linear as $V_{gs} > V_{tn}$ and $V_{out} < V_{in} - V_{tn}$.
- PMOS is in cutoff as $V_{gsp} > V_{tp}$.
- Zero current flows from the supply and so the power dissipation is zero.

Reference: [CMOS Inverter Voltage Transfer Characteristics - VLSI Teacher](#)

Lab Schematics:

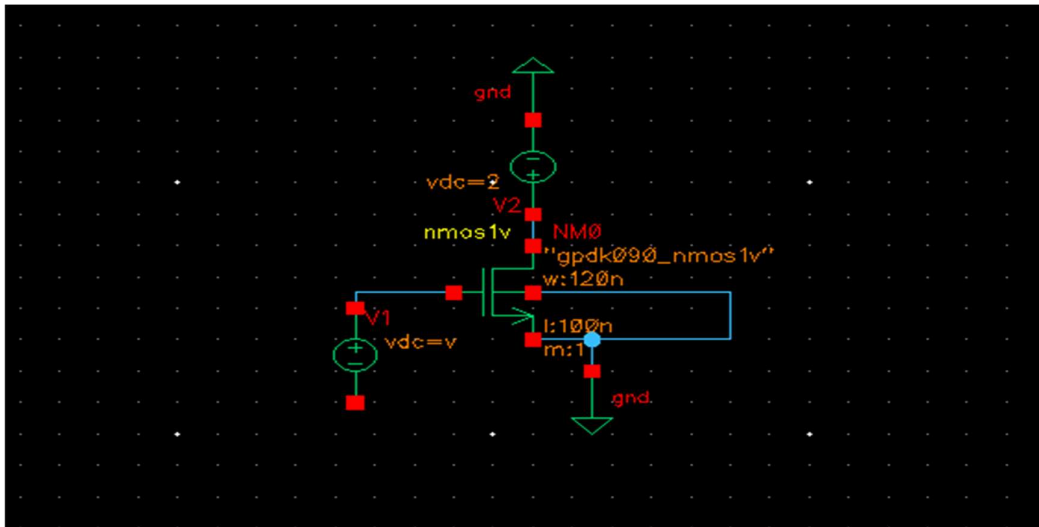


Figure : Schematic of nMOS Transistor

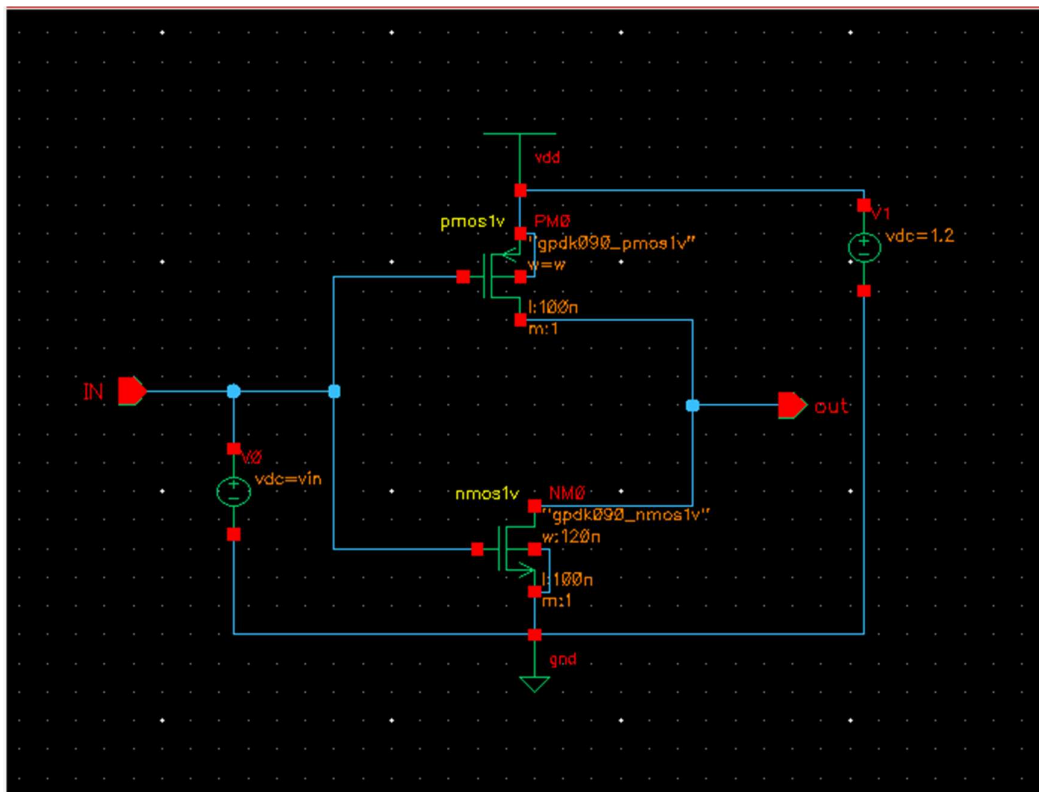


Figure : Setup of CMOS Inverter

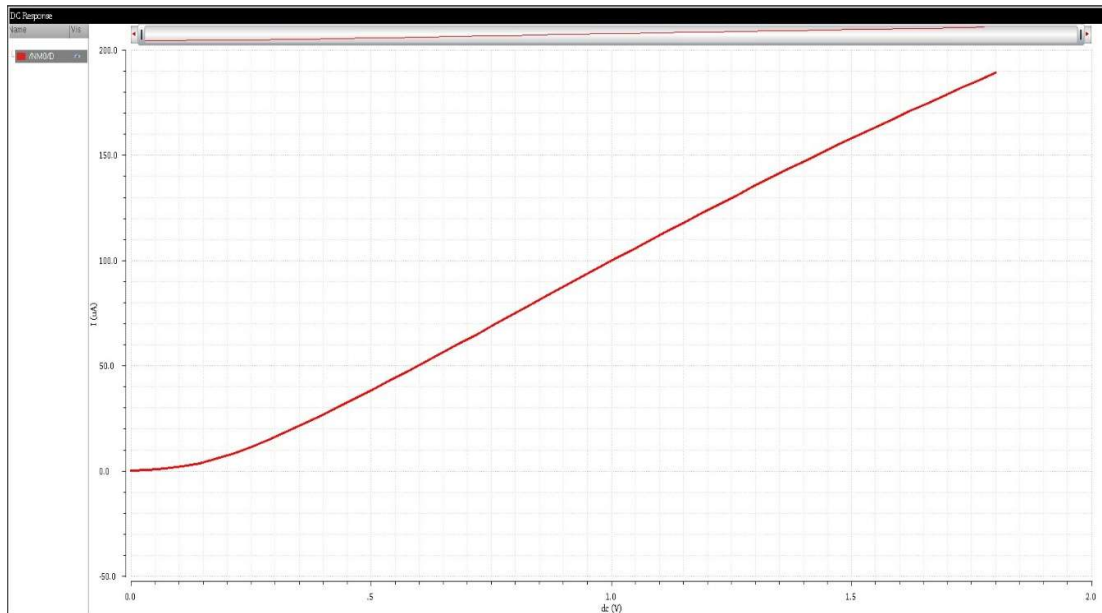


Figure : I_d vs V_{gs}

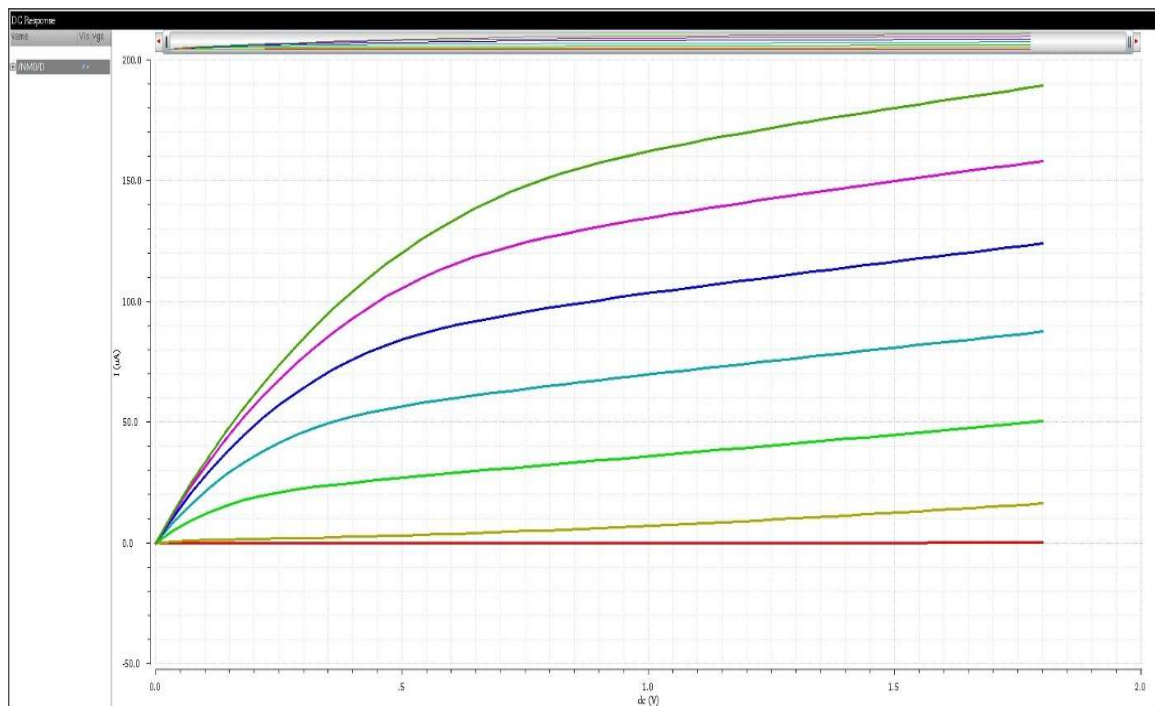


Figure : I_d vs V_{ds} varying V_{gs}

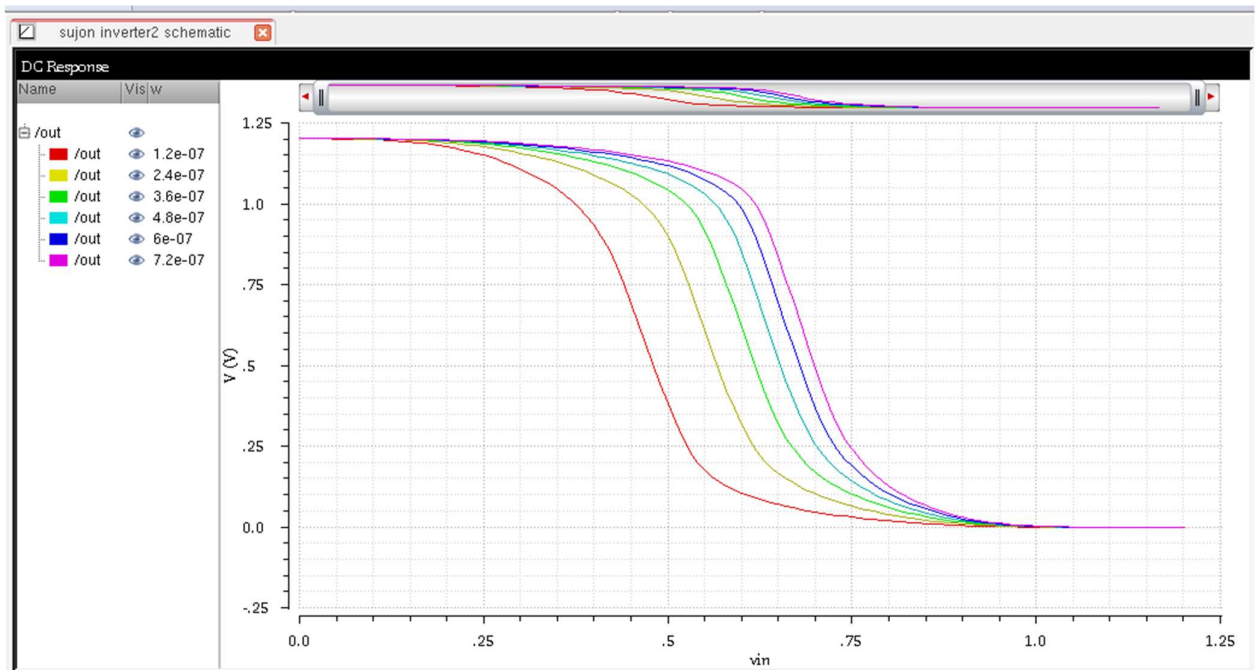


Figure : CMOS Inverter VTC with Varying width (NMOS)

Lab Report 3: Study OF Delay in CMOS inverter And CASCADED INVERTER



Department of Electrical and Electronic Engineering

Course Title: VLSI Design and Testing Lab

Course code: EEE 356

Group No: 12

Submitted By:

Name: Prapto Debnath

Registration No: 2021338016

Name: Sujon Bepary

Registration No: 2021338044

Dept. of Electrical & Electronic Engineering

Submitted To:

Myen Uddin

Lecturer

Department of Electrical & Electronic Engineering,
Shahjalal University of Science and Technology, Sylhet.

Date of Submission

15-08-26

Theory:

CMOS(Complementary Metal-Oxide-Semiconductor) is a widely used technology in digital circuit design that combines both PMOS and NMOS transistors to form efficient logic gates. The most fundamental logic gate built using CMOS is the inverter.

CMOS Inverter Structure: A CMOS inverter consists of one PMOS transistor at the top connected to the supply voltage (VDD) and one NMOS transistor at the bottom connected to ground. Both gates are tied together and act as the input, while the drains of both transistors are connected together to form the output.

Function of CMOS Inverter: When a digital pulse (square wave) is applied at the input:

1 . If the input is low (logic 0), the PMOS turns ON and the NMOS turns OFF. The output is connected to VDD and becomes high (logic1).

2.If the input is high (logic 1), the NMOS turns ON and the PMOS turns OFF. The output is connected to ground and becomes low (logic 0).

Thus, the CMOS inverter provides logical inversion of the input signal.

Timing Parameters: When analyzing dynamic behavior (transient response), three key timing parameters are studied:

Rise Time (trise): The time it takes for the output voltage to rise from 10% to 90% of VDD during a low-to-high transition.

Formula:

$$trise = t_{90\% \text{ level}} - t_{10\% \text{ level}}$$

Fall Time (tfall): The time it takes for the output to fall from 90% to 10% of VDD during a high-to-low transition.

Formula:

$$tfall = t_{10\% \text{ level}} - t_{90\% \text{ level}}$$

Propagation Delay (tpd)

Propagation Delay (tpd): The time taken for the output to respond to a change in the input. This is calculated as the difference between the time at which the input reaches 50% of VDD and the time at which the output reaches 50% of VDD. There are two delays:

– **tpdL:** Delay for a falling output (input rising from 0 to 1)

– **tpdH:** Delay for a rising output (input falling from 1 to 0)

Average propagation delay:

$$t_{pd}(avg) = (t_{pdL} + t_{pdH}) / 2$$

Schematic: CMOS Inverter

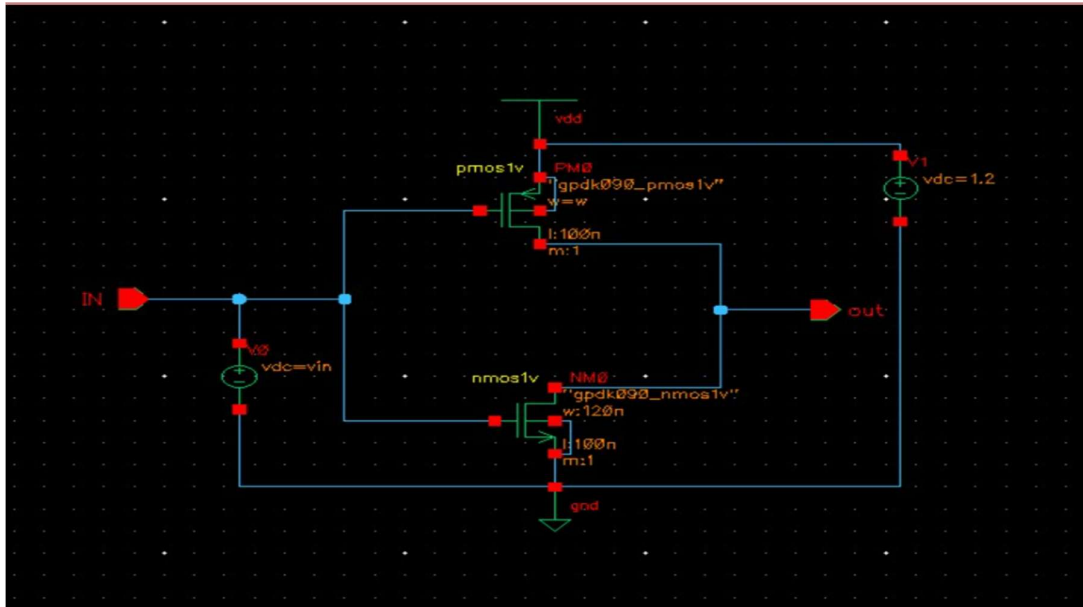


Figure: Circuit schematic of single CMOS inverter

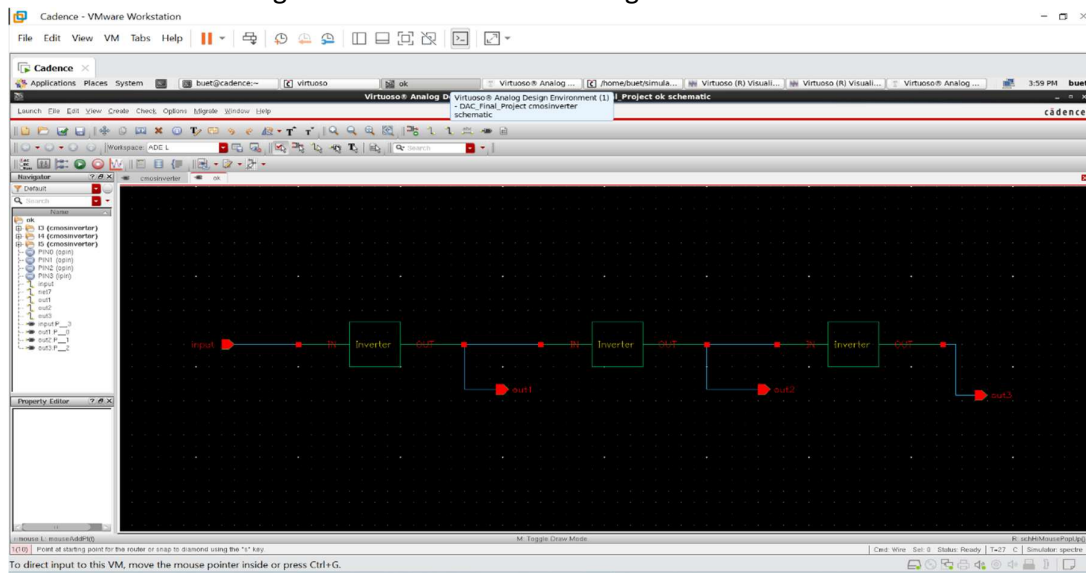
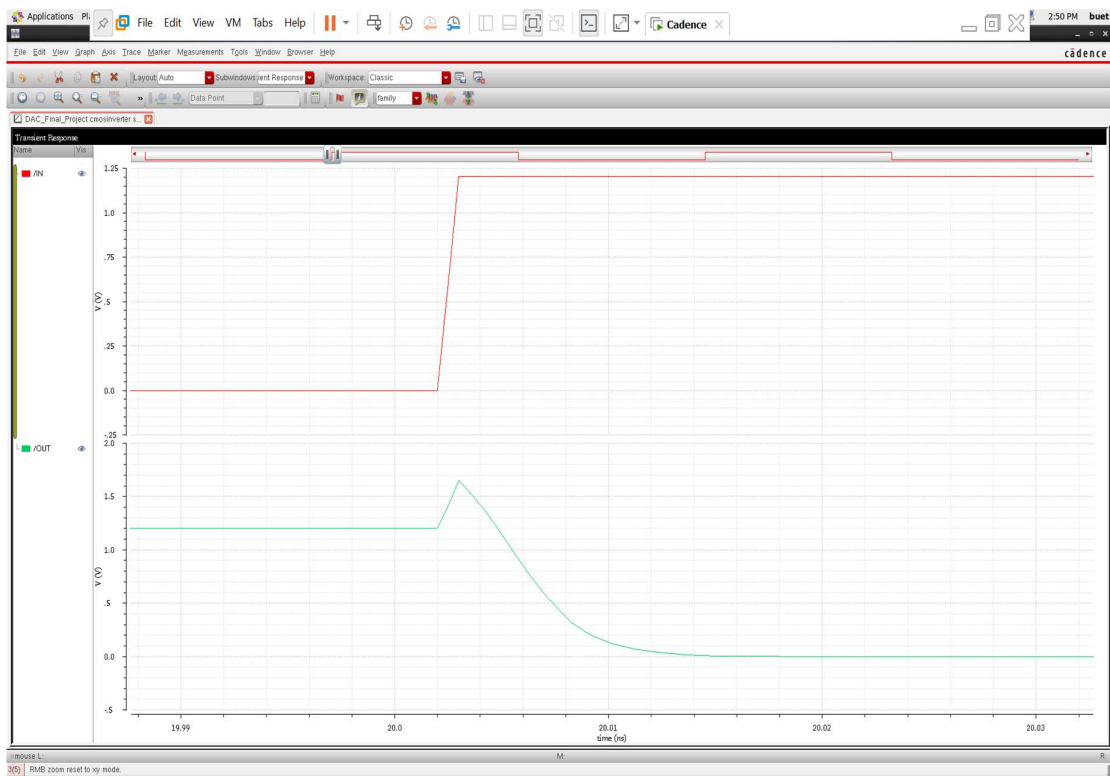
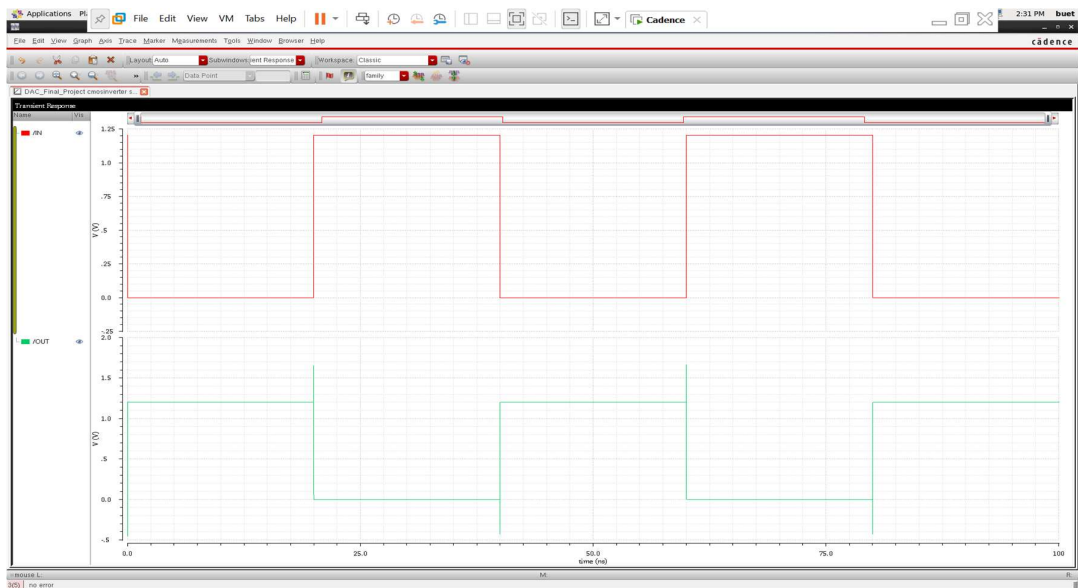


Figure: Circuit schematic of Cascaded CMOS inverter

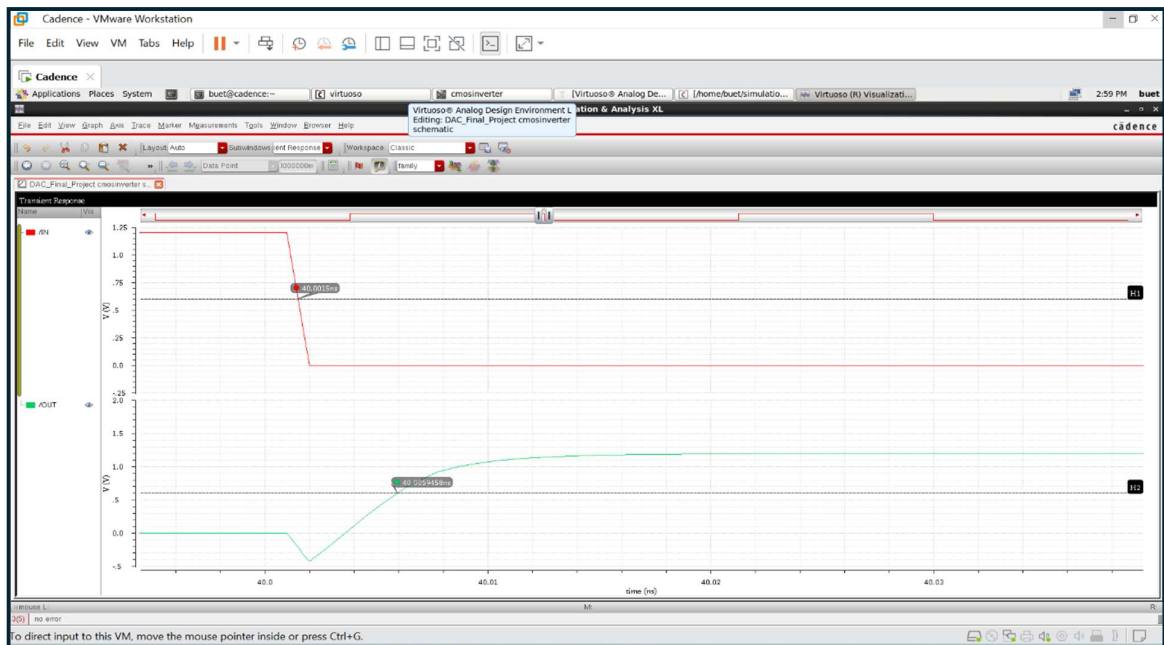
Result :





Delay calculation:

$$\tau_{PHL} = 20.0069355\text{ns} - 20.0025\text{ns} = 0.0044355\text{ns}$$



Delay calculation:

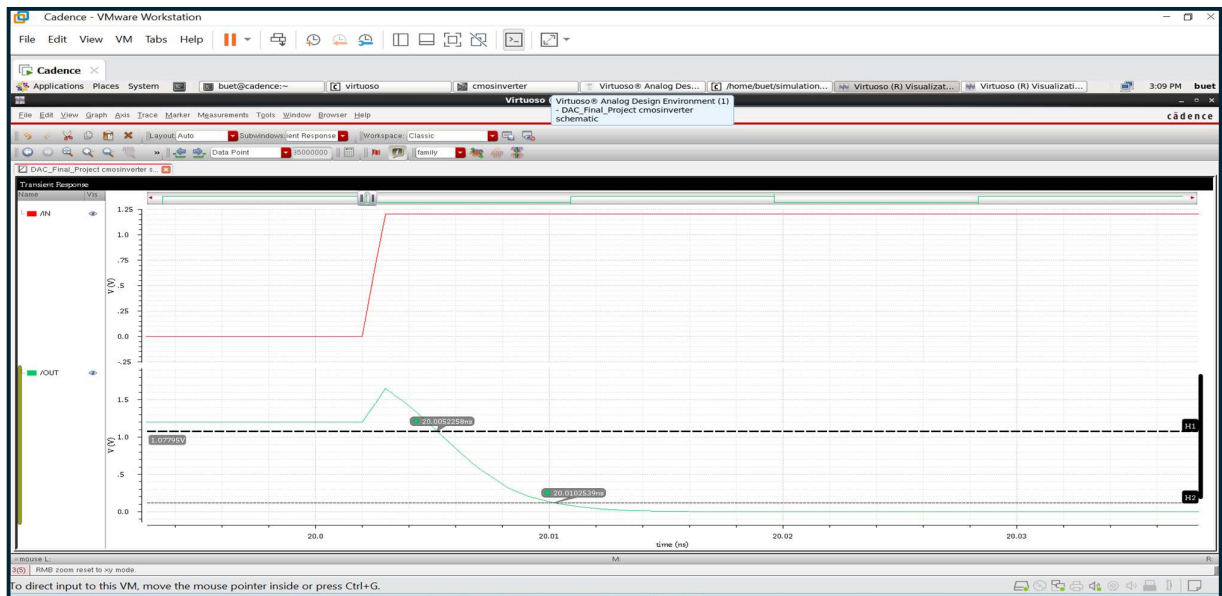
$$\tau_{PLH} = 40.005945\text{ns} - 40.0015\text{ns} = 0.00445\text{ns}$$

Propagation delay:

$$\begin{aligned}\tau_P &= (\tau_{PHL} + \tau_{PLH})/2 \\ &= (0.0044355 + 0.00445)/2 \\ &= 0.00444275\text{ns}\end{aligned}$$

FALL and RISE TIME CALCULATION:

For Fall Time:

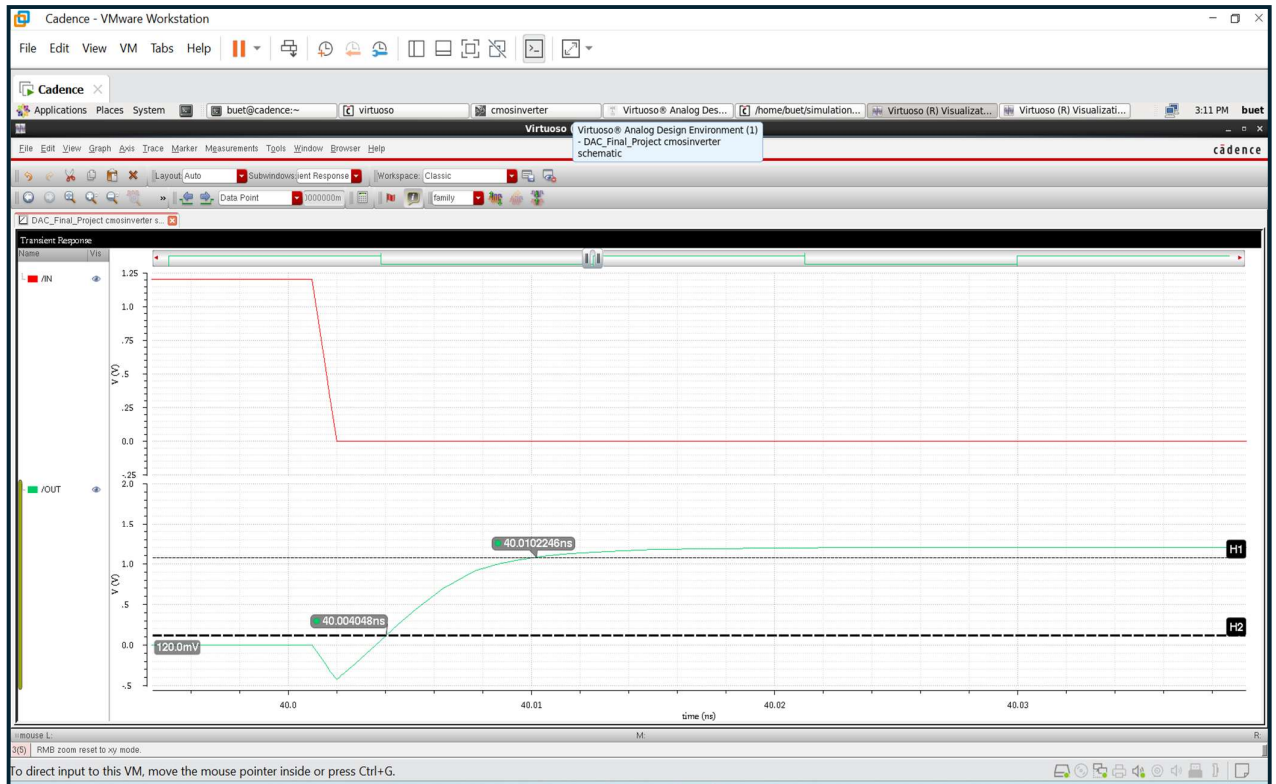


For the calculations here the 90% V_{max} = 1.07982 V and 10% V_{max} = 0.11998 V

$$\begin{aligned}\tau_{Fall} &= 20.010219\text{ns} - 20.00521951\text{ns} \\ &= 0.00499949\text{ns}\end{aligned}$$

This is the fall time

For Rise time:



For the calculations here the 90% Vmax = 1.12 V and 10% Vmax = 0.1238 V

$$\begin{aligned}\tau_{\text{Rise}} &= 40.01022 - 40.004048 \\ &= 0.005974 \text{ ns}\end{aligned}$$

This is the Rise time.

CASCADED INVERTER DELAY:

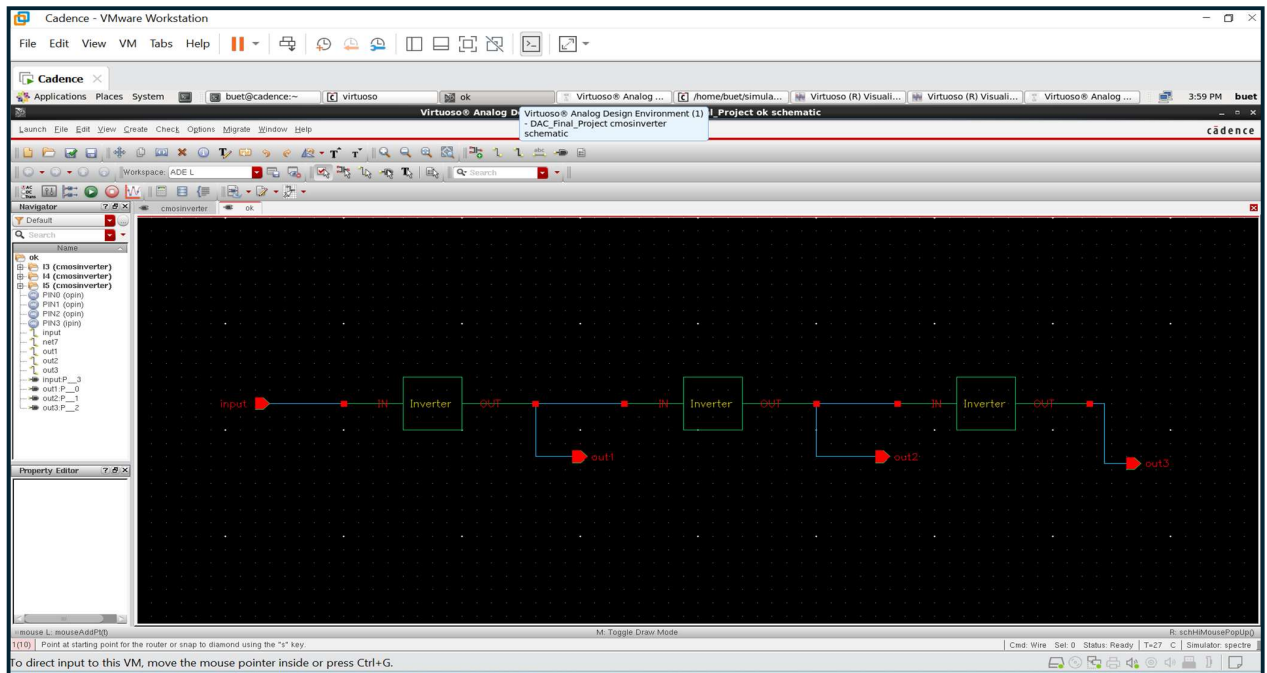


Figure: Schematic of cascaded inverter.

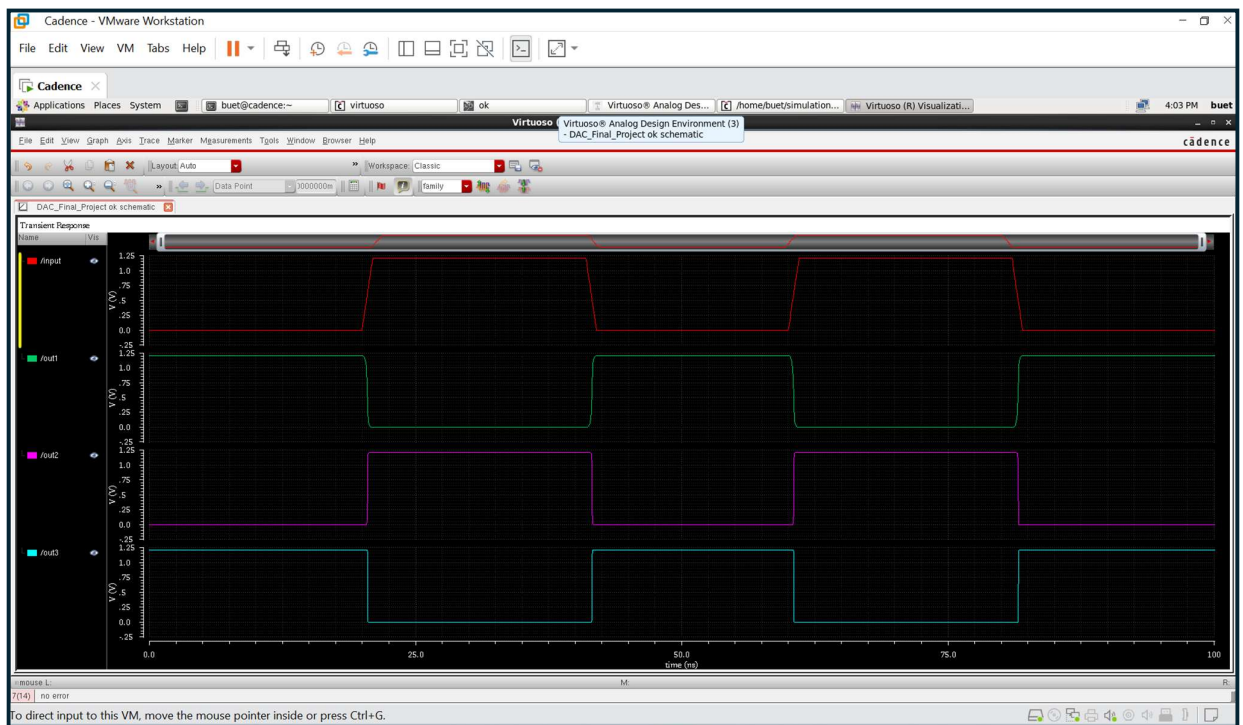


Figure: Output plot of cascaded inverter.

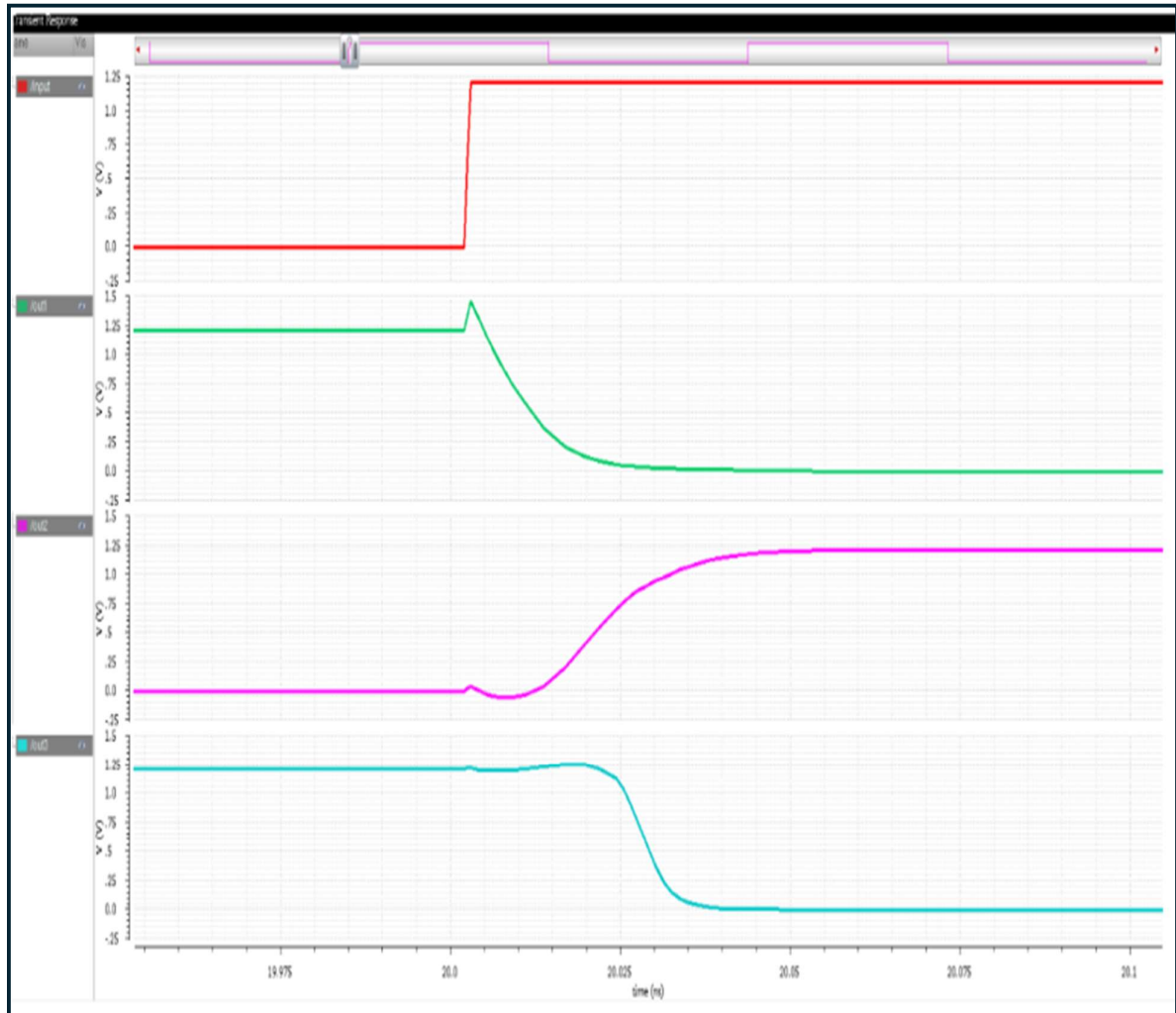


Figure: Output plot of cascaded inverter. (zoomed)

Discussion:

The experiment was performed to study the propagation delay characteristics of a CMOS inverter and a cascaded inverter circuit. From the simulation results, the fall time was 0.00499949 ns, whereas the rise time was 0.005974 ns. The rise time is slightly higher than the fall time because the PMOS and NMOS transistors have different charging and discharging characteristics. The very small rise and fall times indicate that the CMOS inverter can switch between logic states very quickly.

The experiment also shows that when multiple inverters are connected in cascade, the overall propagation delay increases because each inverter stage introduces a certain amount of delay. Therefore, cascading inverter stages helps demonstrate how delay accumulates in multi-stage digital circuits. Overall, the obtained results are consistent with the expected behavior of CMOS circuits and provide a clear understanding of inverter switching delay.

Lab Report 4: Layout Design of an Inverter and Design Validation via DRC



Department of Electrical and Electronic Engineering

Course Title: VLSI Design and Testing Lab

Course code: EEE 356

Group No: 12

Submitted By:

Name: Prapto Debnath

Registration No: 2021338016

Name: Sujon Bepary

Registration No: 2021338044

Dept. of Electrical & Electronic Engineering

Submitted To:

Myen Uddin

Lecturer

Department of Electrical & Electronic Engineering,
Shahjalal University of Science and Technology, Sylhet.

Date of Submission

15-08-26

Theory:

CMOS inverter is the most fundamental building block of digital integrated circuits. It consists of a pMOS and an nMOS transistor connected in a complementary manner. The pMOS transistor is placed at the top and connected to the supply voltage (VDD), while the nMOS transistor is placed at the bottom and connected to ground (GND). Both gates are tied together to form the input terminal, and their drains are connected to form the output.

The operation is simple:

1. When the input is logic '0' (0 V), the pMOS turns ON and the nMOS turns

Off pulling the output to logic '1' (VDD).

2. When the input is logic '1' (VDD), the pMOS turns OFF and the nMOS turns

Off pulling the output to logic '0' (VDD).

This complementary switching action provides full voltage swing, low static power consumption, and high noise margins, making CMOS inverters highly efficient. In VLSI design, the inverter is not only used as a stand-alone gate but also as the basic element for constructing complex logic functions, buffers, and sequential circuits. The layout design in Virtuoso emphasizes proper well formation, diffusion regions, and metal routing to ensure correct physical implementation and performance of the inverter.

LAYOUT:

Layout is representation of a circuit in terms of planar geometric shapes (e.g. rectangles, polygons) showing the patterns of metal, polysilicon, oxide, or diffusion layers that make up the components (resistors, inductors, capacitors, transistors) of the integrated circuit. When using a standard process (e.g. 45nm, 90nm or 180nm process available in our lab), the behaviour of the final integrated circuit depends significantly on the positions and interconnections of the geometric shapes due to parasitic resistances and capacitances contributed by them. While

designing a layout, designer must keep in mind performance (e.g. power-delay product) and size (area occupied by the chip) criterion. While designing digital circuits, one usually follows an ASIC design flow, where the height of standard cells that are used is the same throughout the cell library, but their widths must vary according to their logical functions and drive strengths. The following figure shows a generalized standard cell height .

The most common checks in this verification process are:

1. Design Rule Checking (DRC)

Design Rule Check (DRC):

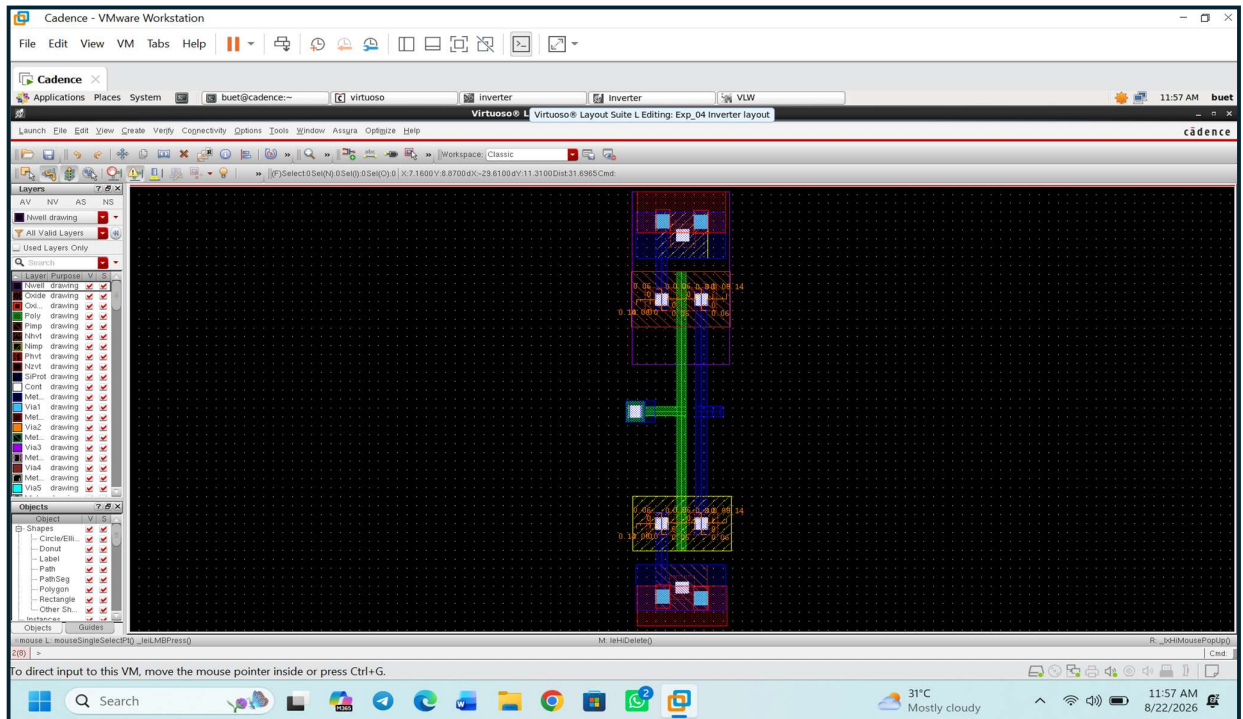
Design Rule Checking (DRC) is the process that determines whether the designed layout of a circuit satisfies a rules specified by the process being used. Design Rules are a series of rules (e.g. area, width, overlap, enclosure, extension, spacing) provided by semiconductor manufacturers which are specific to a particular semiconductor manufacturing process. Design rules specify certain geometric and connectivity restrictions to ensure that the process can fabricate the device properly

Design Rule Checking (DRC) Errors:

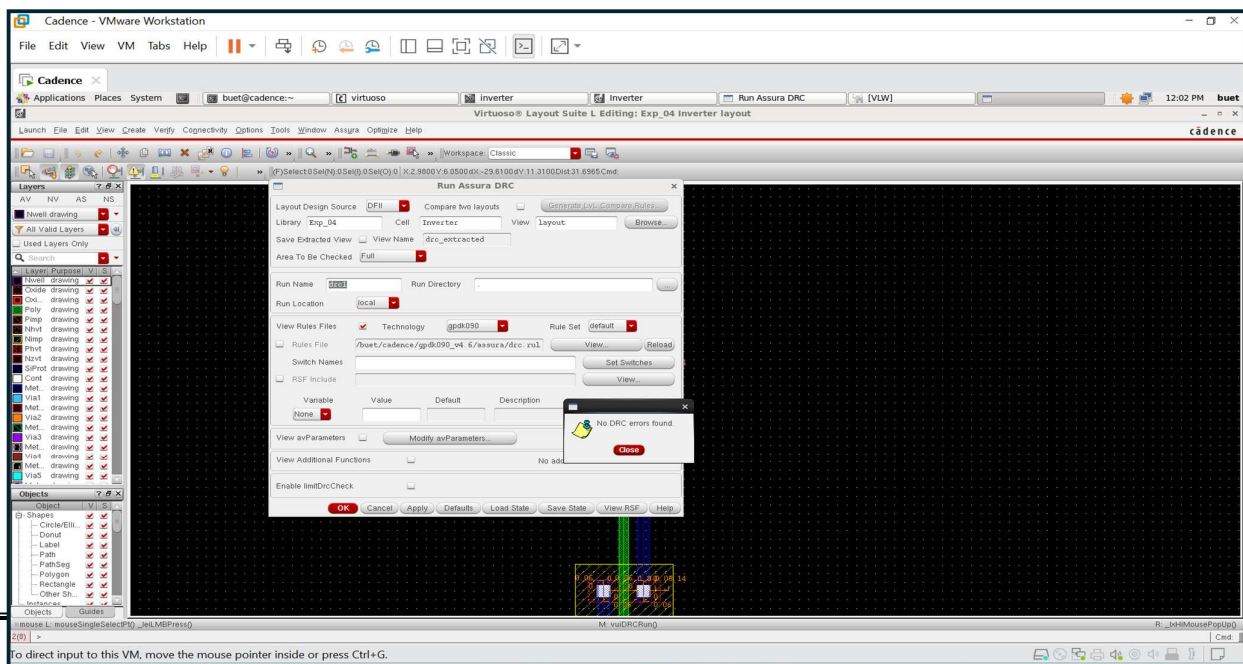
Design Rule Checking (DRC) is an essential verification step in the physical design flow of VLSI circuits. It involves checking the geometrical layout of the integrated circuit against a set of predefined design rules provided by the semiconductor foundry. These design rules ensure that the layout can be manufactured reliably and will function correctly on silicon. DRC errors occur when the layout violates one or more of these rules. Such violations can lead to various fabrication issues, such as short circuits, open circuits, yield loss, or degraded performance. Therefore, DRC plays a crucial role in ensuring design manufacturability and yield. The design rules are based on the limitations of the photolithography and fabrication process. As

technology nodes shrink, these rules become more stringent due to tighter tolerances in device dimensions and spacing.

Schematic:



Result:



Discussion:

In this work, Design Rule Checking (DRC) was carried out on the CMOS layout to verify that the design adhered to all foundry-specified fabrication constraints. The DRC process ensured that minimum width, spacing, enclosure, and area requirements for metal, poly, and diffusion layers were satisfied. Errors such as narrow interconnects, insufficient spacing, and incomplete via enclosures were identified and corrected to achieve a clean layout. Proper well contacts, guard rings, and dummy fills were also added to maintain density balance and device reliability. After resolving all violations, the layout successfully passed DRC verification, confirming that it is fabrication-ready. The process highlights the importance of design compliance, layout accuracy, and rule awareness in physical design. Overall, DRC checking is a critical step in ensuring manufacturability, high yield, and robustness of CMOS circuits before final tape-out.
